

SUPER MARIO BROS PCG & DDA FRAMEWORK

Comprehensive Technical Specification, System Architecture & System Overview

1. Executive Summary & Core Objectives

This project implements a full **Difficulty-Trajectory-Conditioned Procedural Content Generation (PCG) Framework** paired with a **Bayesian Cumulative Dynamic Difficulty Adjustment (DDA) Engine** for Super Mario Bros. The system transitions Mario PCG from single-level generators into a holistic multi-level campaign engine capable of generating contiguous, playable levels tuned precisely to arbitrary designer pacing arcs (e.g. Linear Ramp, Spike & Recovery, Sawtooth Wave, Boss Rush).

Key Features & Innovations:

- **Trajectory-Conditioned 2nd-Order Markov Chain:** Trained on 31 authentic Super Mario Bros VGLC level maps.
- **A* Pathfinding Playability Guarantees:** Integrates Robin Baumgarten A* search solver to validate solvability and enforce a tight difficulty variance constraint ($\leq 5\%$).
- **Selective Cumulative Session DDA Engine:** Evaluates player telemetry across all preceding levels played in the session using exponential decay weighting, adjusting ONLY the next upcoming level.
- **Cumulative Death Zone Telemetry Inspector:** Records and renders ALL death points (X, Y) accumulated across multiple playthrough attempts per level map. Persists across 3-life resets and clears ONLY when brand-new maps are generated.
- **Unified 0–100 Metric Scale:** Harmonizes Chart.js wave-like graphs, designer curve input boxes (5 to 95), telemetry cards, and DDA target metrics onto a single unified 0 to 100% scale.
- **Authentic Classic SMB Level Structures:** Guarantees 8-step staircase intros, 8-step staircase endings with flagpole distance jumps, and automated castle walking sequences.
- **Progressive Enemy Hazards & Stack Block Sanitizer:** Dynamic Green/Red Koopas, Flying Koopas, Spikies, Bullet Bill Cannons, Piranha Plants, and stack block sanitizer guaranteeing 100% hittable reward blocks.
- **Proportional Real-Time Loading Progress Bar:** Modal displaying step-by-step progress (0% to 100%) in exact 1:1 proportion to actual levels compiled and validated.

Module / File		Core Technical Functionality & Responsibility
	pipeline.py	Coordinates full multi-point trajectory generation, DDA simulation, and campaign history aggregation.
	generator.py	Executes conditioned Markov sampling, post-processing decoration enhancements, stack block sanitization, and level generation.
	vglc_trainer.py	Parses 31 VGLC level text files, extracts 2nd-order column transition matrices, and computes difficulty metrics.
	validator.py	Invokes Java Robin Baumgarten A* agent to test solvability and measure structural jump difficulty.
	dda.py	Calculates weighted cumulative session performance indices and enforces bounded micro-adjustments ($\pm 10\%$ max variance).
	game_session.py	Manages player arcade campaign state, life system, coin accumulation, and session progression tracking.
	app.py	Flask web engine providing REST API endpoints for real-time generation, interactive Java play launching, and telemetry polling.
	index.html	Modern glassmorphism Web UI featuring proportional loading progress modal, difficulty curve inspector, tile viewer, and Death Zone overlays.
	PlayLevel.java	Java Swing entry point executing human interactive play and AI playback, returning full telemetry via last_result.txt.

2. Module 1: Trajectory-Conditioned Sequence Generator

Level generation is powered by a 2nd-Order Markov Chain model trained on 31 Super Mario Bros VGLC text levels. The generator conditions contiguous column transitions $P(C_i \mid C_{i-1}, C_{i-2}, D)$ to match designer difficulty vectors $D = [d_1, d_2, \dots, d_N]$.

- **Proportional Loading Bar Modal:** Displays an animated progress bar updating step-by-step (0% to 100%) in exact proportion to actual levels compiled and validated.
- **Authentic Level Intros & Outros:** Every level features an 8-step upward staircase leading to a flagpole placed 4-6 tiles away, followed by an automated walk to a castle door.
- **Unhittable Stack Block Sanitizer:** Includes `_fix_stacked_special_blocks()` to detect reward blocks (1, 2) stacked directly on top of solid obstacles and shift them up to jumping clearance heights or clear them.

3. Module 2: A* Pathfinding Playability Validator

Candidate level grids are evaluated using an automated Robin Baumgarten A* pathfinding search agent. The validator tests if Mario can navigate from start to finish within 200 seconds while tracking jump heights, gap clearances, and hazard avoidance.

- **Strict Solvability Enforcement:** If a candidate level is unplayable or violates target difficulty by >5%, the generator automatically retries up to 50 times.
- **Live AI Playback Window:** Designers can launch a live 3.5x scaled Java window to observe the A* agent solve any generated level in real-time.

4. Module 3: Selective Cumulative Session DDA Engine

The DDA engine dynamically adjusts level difficulty based on cumulative player telemetry across all preceding levels played in the active session. It uses exponential decay weighting ($w_i = 0.70^{n-1-i}$) to combine overall session trend with recent performance.

- **Selective Next-Level Tuning & Slot Retention:** On course clear, DDA pre-adjusts the upcoming level $k+1$. On player death, the session strictly remains locked to Level slot k with targeted spatial DDA micro-adjustments (pit platform injection, enemy demotion, power-up insertion).
- **Strict 10% Variance Bound:** Bounds difficulty adjustments to $\pm 10\%$ of the designer's target curve.
- **Automatic Session Reset:** Generating a new campaign sequence automatically clears cumulative DDA session history and resets active DDA shifts to 0.0%.

5. Module 4: Visualizer, Death Zones & Life System

- **Bounded Horizontal Scrolling Visualizer:** Locks tile view to grid height (260px) with custom glowing horizontal scrollbars spanning all 220 columns.
- **Synchronized Canvas Coordinate Space:** Wraps `#tileGrid` and `#heatmapCanvas` in 1:1 pixel coordinate alignment across all 220 columns for death markers (■) and AI playback.
- **Cumulative Death Zone Telemetry:** Records and renders ALL death points (X, Y) accumulated across multiple attempts/retries for each level map. Death markers persist across 3-life campaign resets and are ONLY cleared when brand-new maps are generated.
- **Unified 0–100 Metric Scale:** Harmonizes Chart.js wave-like graphs, designer curve input boxes (5 to 95), telemetry cards, and DDA target metrics onto a single unified 0 to 100% scale.
- **3-Life System & Game Over Guard:** Clamps lives at 0 on death; on 3rd death, halts play with Game Over state and disables the Play button until 'Reset Arcade Progression' is clicked.
- **In-Game Swing HUD Sync:** Passes player lives and coins into Java MarioWorld, cleanly rendering remaining lives on the top-left in-game HUD.

6. Project Remarks & Audit Logging System

All updates and architecture changes are systematically recorded in the Remarks/ directory with date-stamped logs (e.g. Remarks/2026-08-06.txt). This maintains a transparent, complete historical record of project progress.
